Lab Practice: Implementing a Decision Tree with Different Splitting Criteria

Objective

Implement a Decision Tree classification algorithm using two splitting criteria: **Gini Impurity** and **Entropy / Information Gain**.

Instructions

1. Use the provided categorical dataset to build a Decision Tree classifier.
2. Implement the splitting criterion using **Gini Impurity**.
3. Implement the splitting criterion using **Entropy and Information Gain**.
4. At each node, evaluate the available features and select the best feature for splitting based on the chosen criterion.
5. Construct the decision tree recursively until the stopping conditions are met.
6. Test both implementations using the same dataset and compare the resulting tree structures and classification predictions.

Requirements

- Use Python to implement the algorithms.
- All input features must be categorical.
- Do not use an existing Decision Tree implementation, such as `sklearn.tree.DecisionTreeClassifier`, to build the tree.
- You may use libraries such as Pandas and NumPy for data handling and calculations.
- Submit your source code and a short explanation comparing the two splitting criteria.

Expected Learning Outcome

Understand how Gini Impurity and Entropy / Information Gain select features for splitting and how different criteria may produce different Decision Tree structures.